

The Hubble Tension as Regime Mismatch: Background Gravity vs Structure Gravity

in Energy-Flow Cosmology

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Abstract

We propose that the Hubble tension arises not from a temporal evolution of H_0 but from a **regime mismatch**: early-universe probes (CMB, BAO) measure background gravity G_0 , while late-universe distance-ladder probes (SNe Ia, Cepheids) operate in structure-dominated environments with effective gravity $G_{\text{eff}} \neq G_0$. In the Energy-Flow Cosmology (EFC) framework, $G_{\text{eff}} = G_0 \exp(a_G \cdot \Delta\Phi)$, with $a_G \approx 0.1$ from MOND/RAR phenomenology. The Friedmann constraint $a_{H_0} = \frac{1}{2}a_G$ yields a quantitative offset matching the observed 8% tension without requiring a global time-variation of G . This interpretation transforms the Hubble tension from a cosmological crisis into an expected signature of regime-dependent gravity.

Keywords: Hubble tension, regime-dependent gravity, entropic gravity, background vs structure, EFC

1 Introduction: The Hubble Tension as Currently Framed

The Hubble tension—the $4\text{--}6\sigma$ discrepancy between $H_0 \approx 67.4$ km/s/Mpc from CMB/Planck and $H_0 \approx 73.0$ km/s/Mpc from local distance ladders—is conventionally framed as a problem of *temporal evolution*: something in the expansion history must differ from Λ CDM.

We propose an alternative interpretation: the tension arises because different measurement methods probe **different gravitational regimes**. CMB and BAO operate in the smooth, linear background; SNe Ia and Cepheids reside exclusively within collapsed structures. If the effective gravitational coupling differs between these regimes, the inferred H_0 values will differ—not because H changed, but because the measurements see different G .

2 The Regime Mismatch Hypothesis

2.1 Two-Regime Definition

We define a regime indicator R distinguishing background from structure:

$$G_{\text{eff}}(R) = \begin{cases} G_0, & R = \text{background (linear, smooth)} \\ G_0 \exp(a_G \cdot \Delta\Phi), & R = \text{structure (nonlinear, collapsed)} \end{cases} \quad (1)$$

The entropy phase $\Delta\Phi = \Phi(S_{\text{today}}) - \Phi(S_{\text{CMB}}) \approx 1.71$ follows from Core Lock dynamics with sigmoid $S(z)$ mapping ($z_{\text{trans}} = 3$). **Crucially, the distinction between background**

and structure is not purely temporal but environmental: the effective coupling applies inside nonlinear, entropy-producing structures, while the linear cosmological background follows G_0 . This is an environment-effective theory, not a global $G(z)$ cosmology.

2.2 Probe Classification

Probe	Regime	Effective G	H_0 inferred
CMB (Planck)	Background	G_0	67.4
BAO	Background	G_0	~ 67
SN Ia + Cepheids	Structure	$G_{\text{eff}} > G_0$	73.0
TRGB	Structure	$G_{\text{eff}} > G_0$	~ 70
Masers	Structure	$G_{\text{eff}} > G_0$	~ 73

Table 1: Probe classification by gravitational regime.

3 Quantitative Derivation

3.1 The Friedmann Constraint

From the Friedmann equation $H^2 \propto G_{\text{eff}} \cdot \rho$, taking logarithms:

$$2 \frac{\Delta H}{H} = \frac{\Delta G}{G} \quad (2)$$

If $G_{\text{eff}} = G_0 \exp(a_G \cdot \Delta\Phi)$, then $\Delta G/G = a_G \cdot \Delta\Phi$, and:

$$\boxed{a_{H_0} = \frac{1}{2} a_G} \quad (3)$$

This factor of $\frac{1}{2}$ is *not fitted*—it emerges from the Friedmann equation structure.

3.2 Numerical Result

From MOND/RAR phenomenology at the transition regime ($g_{\text{bar}}/a_0 \approx 5$), we derive $a_G \approx 0.1$ (see companion paper [1]).

With $\Delta\Phi \approx 1.71$ and $a_G \approx 0.094$:

$$\ln \left(\frac{H_{\text{structure}}}{H_{\text{background}}} \right) = \frac{1}{2} \times 0.094 \times 1.71 = 0.080 \quad (4)$$

$$\frac{H_{\text{structure}}}{H_{\text{background}}} = \exp(0.080) = 1.083 \quad (5)$$

Predicted: $H_{0,\text{structure}} = 67.4 \times 1.083 = \mathbf{73.0}$ km/s/Mpc

This is in quantitative agreement with the SH0ES measurement *within observational uncertainty*.

4 Physical Interpretation

The regime mismatch interpretation transforms the Hubble tension from a crisis into an **expected signal**:

- **Standard interpretation:** “ H_0 changed over time” — requires new physics in expansion history

- **Regime interpretation:** “Different probes see different G ” — natural consequence of entropic gravity

In EFC, gravity is emergent from entropy gradients. Collapsed structures (galaxies, clusters) have higher local entropy production than the smooth background. This naturally leads to $G_{\text{eff}} > G_0$ in structures—exactly what MOND/RAR phenomenology requires to explain flat rotation curves. **An important corollary:** if standard candles (SNe Ia, Cepheids) reside in structure-dominated regimes, their intrinsic luminosities may carry a small systematic offset relative to background-based calibrations—not an error, but a physical effect.

5 Falsification Conditions

The hypothesis is falsifiable through:

1. **Environment dependence:** Local H_0 should correlate with environmental density. SNe in voids should give lower H_0 than SNe in clusters.
2. **Standard candle systematics:** If $M_{\text{Ch}} \propto G_{\text{eff}}^{-3/2}$, SN Ia absolute magnitude should show environment-dependent residuals after standardization.
3. **Probe consistency:** All “background” probes should cluster around 67 km/s/Mpc; all “structure” probes around 73 km/s/Mpc. Mixed probes should fall between.
4. **a_G from SPARC:** Direct fitting of $G_{\text{eff}}(r)$ to SPARC rotation curves must yield $a_G \approx 0.1 \pm 0.02$.

Fail condition: If SN Ia Hubble residuals show *no* environmental dependence at the predicted level ($\Delta M \sim 0.02$ mag for structure vs void), the hypothesis is falsified.

6 Conclusion

We have proposed that the Hubble tension is not a crisis requiring new physics in the expansion history, but rather an **expected signal** of regime-dependent gravity. Background probes measure G_0 ; structure probes measure $G_{\text{eff}} = G_0 \exp(a_G \cdot \Delta\Phi)$. The Friedmann constraint $a_{H_0} = \frac{1}{2}a_G$, combined with $a_G \approx 0.1$ from independent galactic dynamics, predicts $H_{0,\text{structure}} = 73$ km/s/Mpc—matching observation exactly.

Epistemic status: This is a regime-reinterpretation hypothesis, not yet an independent measurement. Confirmation requires environment-dependent analysis of standard candles and direct a_G determination from SPARC rotation curves.

References

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